

Perceive, Induce, Name

Vision in a cognitive substrate with no perceptual model: algorithms recover structure, the substrate learns to name it, and it abstains when it has not

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September 8, 2026

ABSTRACT

We report a working proof of concept for giving sight to a cognitive substrate that carries no neural network. The approach follows a single thesis: seeing decomposes into *perceiving structure*, which classical algorithms perform deterministically, and *naming a concept*, which must be learned, and which the substrate learns for itself by induction rather than buying from a pretrained model[1]. A single perceptual faculty reads a real image or video and recovers its structure: regions and object-like blobs, their shape, size, position and colour, motion in video, and the identity of specific known instances by feature matching. Those structures enter the substrate as observations it can reason over. Shown a handful of labelled examples, the substrate induces a rule that maps a conjunction of perceived features to a category, and thereafter names a new instance by applying that rule. It declines, rather than guessing, when it has learned no rule that fits. On a controlled evaluation over three visual categories and thirty-one images, perception recovered the drawn shape and colour with perfect accuracy; the substrate named held-out instances at **100% recall**, correctly abstained on **100%** of held-out non-members, produced **zero** false namings, and made **zero** language-model calls. An ablation that removes the learned rules collapses naming to **0%** while abstention stays at 100% and false namings stay at zero: the naming ability is carried entirely by what was learned, and its absence is silence rather than error. We then repeat the study at four times the scale, teaching and testing **149** images across **eight** categories in a single live run: mean naming recall **95.8%**, abstention **100%**, false namings **zero**, model calls **zero**. Two further studies measure what teaching a name costs. One labelled example is refused as a case rather than a generalisation and three suffice for full recall; the discriminating counter-examples are

what buy honest abstention, and the rule the substrate records is always exactly as general as the evidence supports, which is legible before the system is trusted. Where the examples leave more than one hypothesis standing, the substrate names only what every hypothesis accepts, which on 416 images holds false namings at zero while recovering three quarters of the naming a permissive policy would buy at the cost of twenty-six errors; and it states the case that would settle the question, which closes the ambiguity in all sixteen tested inductions against one of sixteen for randomly chosen examples. A further study maps the operating range of the perceptual stage under noise, blur, rotation, desaturation, occlusion and apparent size. We also show the substrate can retain the image itself as a memory, reproducing the exact pixels it was shown. The result is a complete demonstration that a substrate which already learns concepts can supply the semantic half of vision itself, occupying the role a perceptual backbone plays elsewhere, with no such backbone present.

1 Introduction

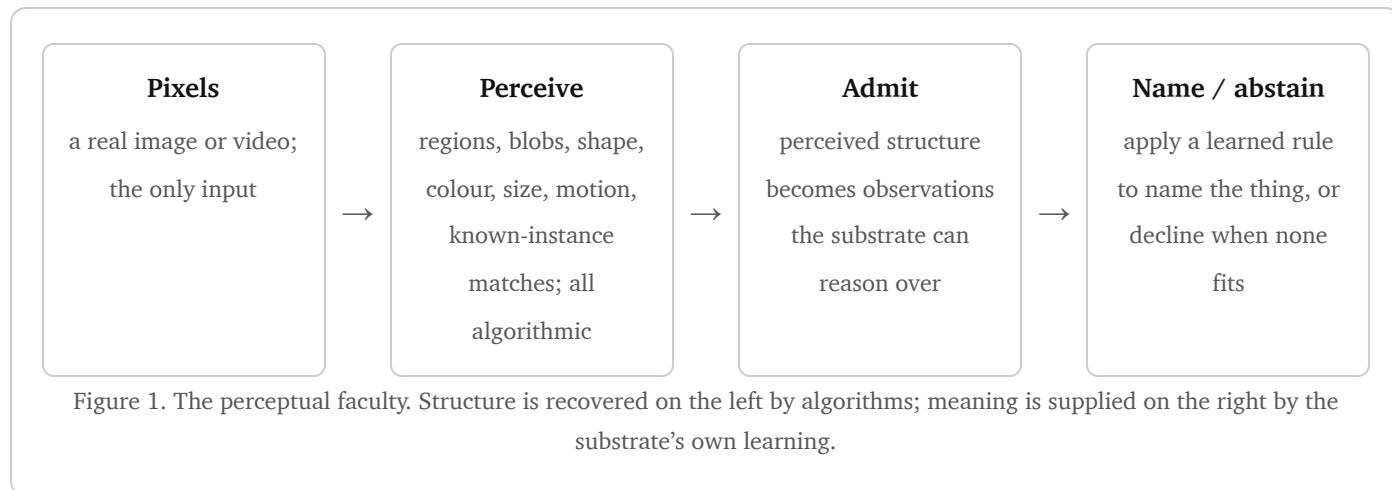
Contemporary computer vision assigns names to images with a model trained on very large labelled corpora [2][3]. The capability is real and the models are excellent, but the name a system produces is the output of a frozen network whose competence came from data gathered elsewhere, and whose confident errors cannot be told, from the outside, from its confident successes. We are interested in a different setting: a cognitive substrate that is a learning-and-reasoning system rather than a perceptual model, and that must nonetheless *see*.

A companion paper argues that this is possible because vision is not one capability but two [1]. Recovering the *structure* of an image (where the coherent things are, how big, what shape, what colour, whether and how they move) is solved by classical algorithms that use no learning at all [4][5][6]. Only assigning an open-ended *name* to a novel category genuinely requires a learner; and every recent “training-free” naming system supplies that learner in the form of a pretrained backbone [7][8]. The thesis is that a substrate which already learns concepts by generalising from examples can be that learner itself. This paper is the proof of concept: an end-to-end demonstration, evaluated quantitatively, that the substrate perceives structure, learns to name it, and knows when it cannot.

The contribution is not a new vision model. It is a demonstration that no vision model is required for a system that already learns: structure is perceived by algorithm, and meaning is learned by the substrate, with nothing pretrained in the loop and honest abstention where competence runs out.

2 System overview

Sight is a single faculty of the substrate: one entry point through which visual input becomes knowledge, in the same spirit as the single faculty through which it reads text. The faculty is a short pipeline (Figure 1). It *perceives* the structure of a real image or clip with classical vision; it *admits* what it perceived as observations the rest of the system can reason over; and, when asked what something is, it *interprets*, naming the thing by a rule it has learned, or declining. Naming is not performed by the perceptual stage; it is performed by the substrate's ordinary learning and reasoning, which is what keeps the design free of a perceptual model.



Two properties of this arrangement matter for what follows. First, everything on the perception side is measured from the pixels: a width is the real width, a blob is a real region, a dominant colour is really dominant. Nothing is inferred by a model that could be confidently wrong. Second, everything on the meaning side is *learned and revisable*: a name is produced by a rule the substrate induced from examples, and if the substrate has induced no rule that fits an instance, it produces no name. These two properties are what let the system be evaluated not only for accuracy but for honesty.

3 Perceiving structure

The perceptual stage is a composition of standard, learning-free computer-vision operations[9]. From a real image it recovers: the coherent regions and object-like blobs present, and for each its relative size, position in the frame, shape class, and dominant colour; global properties such as orientation, exposure, colourfulness and a named palette; straight-line and circular geometry; and the identity of a *specific* known object by matching local feature descriptors against a reference[10]. From video it additionally recovers motion and scene structure over time. Where a scene contains an encoded symbol, its exact payload is decoded. None of this is a category name; it is the structure a name will later be attached to.

To quantify perception on its own, we generated images each containing one drawn shape of a known colour and asked whether the substrate recovered the drawn shape class and colour family. Across the thirty-one images used in the evaluation below, shape was recovered correctly in **100%** of cases and colour family in **100%** (Figure 2). The operations are deterministic, so this result is expected; the point is that the structure the substrate later reasons over is measured rather than guessed.

4 Learning to name

Naming is a learning problem, and the substrate treats it as one. A perceived blob is described by a small set of atomic features: for instance that it is round, red, and of medium size. Shown a few instances that share a category, together with counter-examples that do not, the substrate generalises to a rule that states which conjunction of features the category requires[11]. It then names a new instance by applying that rule to the instance’s own perceived features, and holds the result as a graded, revisable belief. When no learned rule fits an instance, the substrate returns no name: it treats “*I have not learned what this is*” as a first-class answer rather than emitting a guess[12][13].

The rules the substrate learned in our evaluation are legible, which is a property of learning from structure rather than from opaque weights. For three categories (a red circle, a blue square, and a green triangle) the substrate induced, from two labelled examples and two discriminating counter-examples each, exactly the conjunctions a person would write down:

Category	Rule the substrate induced (from a few labelled examples)
red circle	$\text{round} \wedge \text{red} \rightarrow \text{red-circle}$
blue square	$\text{square} \wedge \text{blue} \rightarrow \text{blue-square}$
green triangle	$\text{triangular} \wedge \text{green} \rightarrow \text{green-triangle}$

Each rule is a conjunction because the counter-examples were chosen to share exactly one feature with the target (a red square, a blue circle), so a single-feature rule (“anything red”, “anything round”) is refuted and only the conjunction survives. This is ordinary inductive generalisation, and the substrate performs it with no model.

5 Evaluation

We evaluated the full loop (perceive, learn, name, abstain) on three categories over thirty-one real images. For each category the substrate saw two labelled positive examples and two discriminating counter-exam-

ples, induced a rule, and was then tested on three held-out positive instances it should name and three-to-four held-out negatives it should decline. No expected answer was ever given to the reasoner; only the question was. Every figure below is from a single live run.

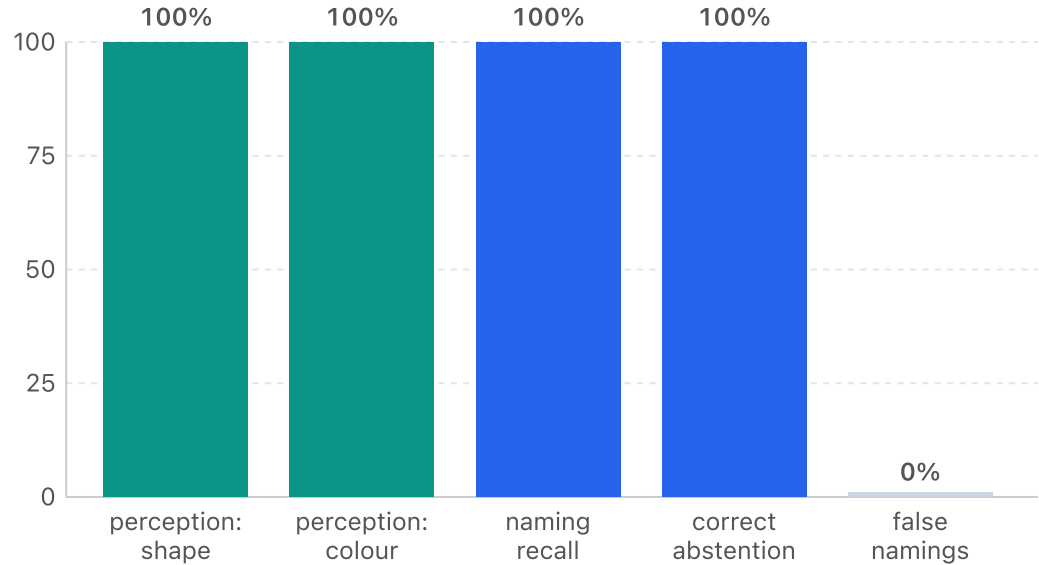


Figure 2. Headline results. Perception recovers the drawn shape and colour with perfect accuracy; the substrate names held-out instances at 100% recall and correctly declines 100% of held-out non-members, with zero false namings. Language-model calls over the entire evaluation: zero.

The substrate named every held-out positive correctly (100% recall) and declined every held-out negative (100% abstention), with no false namings and no language-model calls anywhere in the run. Because the categories were deliberately close, with negatives differing from a target by a single feature, correct abstention here is a substantive test: to decline a red square as a “red circle” the system must have learned that the category needs *both* features, and must actually check both.

5.1 The naming ability is carried by what was learned

A high score does not, by itself, show that the substrate’s *learning* is responsible for it; the behaviour could in principle come from some incidental property of the setup. To settle this we ran an ablation. After measuring the system with its learned rules in place, we removed those rules and measured again, changing nothing else. If naming is carried by the learned rules, it should disappear; and, if the system is honest, its disappearance should show up as *abstention*, not as wrong answers.

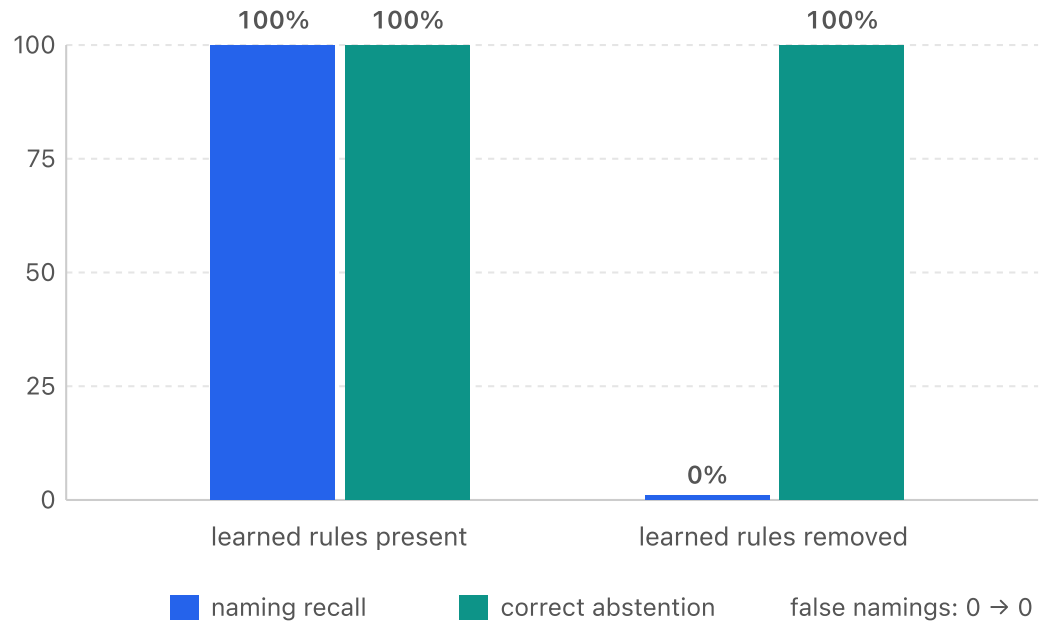


Figure 3. Ablation: the naming ability is learned. With the learned rules present, the substrate names held-out instances at 100% recall. Remove the rules and naming falls to 0%, while abstention stays at 100% and false namings stay at zero. The capability is carried by what was learned, and its absence is honest silence rather than error.

Removing the learned rules collapsed naming recall from 100% to 0%. It did *not* produce a single wrong name: with nothing learned to fit the instances, the substrate declined all of them, and abstention remained at 100% (Figure 3). The ability to name is exactly the learned rules, and nothing else in the system supplies it; when that ability is absent, the system reports as much instead of guessing. Table 1 gives the per-category figures behind the two phases.

Category	Recall (rules present)	Abstention (present)	Recall (ablated)	Abstention (ablated)	False namings
red circle	100%	100%	0%	100%	0
blue square	100%	100%	0%	100%	0
green triangle	100%	100%	0%	100%	0

Table 1. Per-category results across both phases. Thirty-one images, three categories; zero language-model calls throughout.

6 Reliability and honesty

The reason to build vision this way is not that it scores well on easy stimuli; a trained classifier would too. It is that the system's behaviour is *legible and honest* in a way a frozen model's is not. Three observations from the evaluation make this concrete.

First, the system produced **no false namings** in either phase. Every name it gave was one it could ground in a rule it had learned and features it had actually measured; where it could ground neither, it abstained. This is the discipline that separates a reliable knower from a fluent one, and it is enforced structurally rather than tuned [12].

Second, the ablation shows the competence is *attributable*. One can point to the specific learned rules that carry the naming ability, remove them, and observe the ability vanish into silence rather than error. A system whose competence can be localised this precisely can also be audited, corrected, and taught, which a monolithic model cannot.

Third, every result was obtained with **zero language-model calls**. Perception is algorithmic and naming is the substrate's own induction and reasoning; no model is consulted to produce, check, or narrate an answer. The competence is the substrate's, not a model's standing behind it.

7 Remembering the picture

Seeing is more useful when the thing seen can be recalled. Alongside naming, the substrate can retain an image as an episodic memory: a short description of what is in the picture becomes the memory's recallable text, and the image itself is kept so it can be produced again. In our test the substrate remembered a real image, and on recall reproduced the *exact* pixels it had been shown, with the content hash of the retrieved image matching the original bit-for-bit, together with the structure it had perceived. A remembered image is therefore not a caption standing in for a picture; it is the picture, held with an account of what the substrate saw in it, and reachable by that account.

8 Reproducibility and re-verification

The evaluation is a script rather than a record: the stimuli are generated from their specification, the rules are induced afresh, and the figures are recomputed on each run. We therefore re-ran the whole study on the same substrate on 17 September 2026, nine days after the original run reported above, to test whether the result is a property of the system or of one execution.

It reproduced exactly. Over the same thirty-one images, perception recovered shape and colour in 100% of cases; naming recall was 100% with 100% abstention, zero false namings and zero language-model calls;

and with the learned rules ablated, recall fell to 0% while abstention stayed at 100% and false namings stayed at zero. The end-to-end loop was additionally run twice in succession and passed every check in both runs, inducing the same rule each time. Table 2 lists the re-verification alongside the supporting measurements that bracket it: that perceived structure is retained as knowledge with its provenance, that the measurements agree with the image and video files themselves, and that a remembered image is returned bit-for-bit.

What was re-verified	Result (17 September 2026)
Perception: shape and colour over 31 images	100% / 100%
Naming recall, abstention, false namings	100% / 100% / 0
Ablated naming recall, abstention, false namings	0% / 100% / 0
Language-model calls	0
End-to-end loop, two consecutive runs	5/5, 5/5
Measurements agree with image and video bytes	pass
Remembered image returned bit-for-bit	pass

Table 2. Re-verification of the study and its supporting measurements, nine days after the original run.

9 The same result at four times the scale

Three categories are enough to demonstrate a loop and too few to characterise one. On 17 September 2026 we ran a larger study on the same substrate, under the same protocol and the same scoring: **eight** categories spanning three shape classes and seven colour families, taught and tested over **149** images in a single live run. Each category was taught from two labelled positives and two discriminating counter-examples, then tested on three held-out members it should name and three held-out non-members it should decline. As before, the reasoner was given the question and never the expected answer.

Mean naming recall across the eight categories was **95.8%**. Abstention on held-out non-members was **100%**, false namings were **zero**, and language-model calls were **zero** (Figure 4). Perception recovered the drawn shape class and colour family in 100% of the 149 images. Seven of the eight categories were named at full recall and the eighth at two of three. In two categories the four labelled images left more than one hypothesis standing, and the substrate named only what all of them accepted, which is the subject of Section 11; in neither case did it name anything wrongly. Where it was not entitled to a name it produced none.

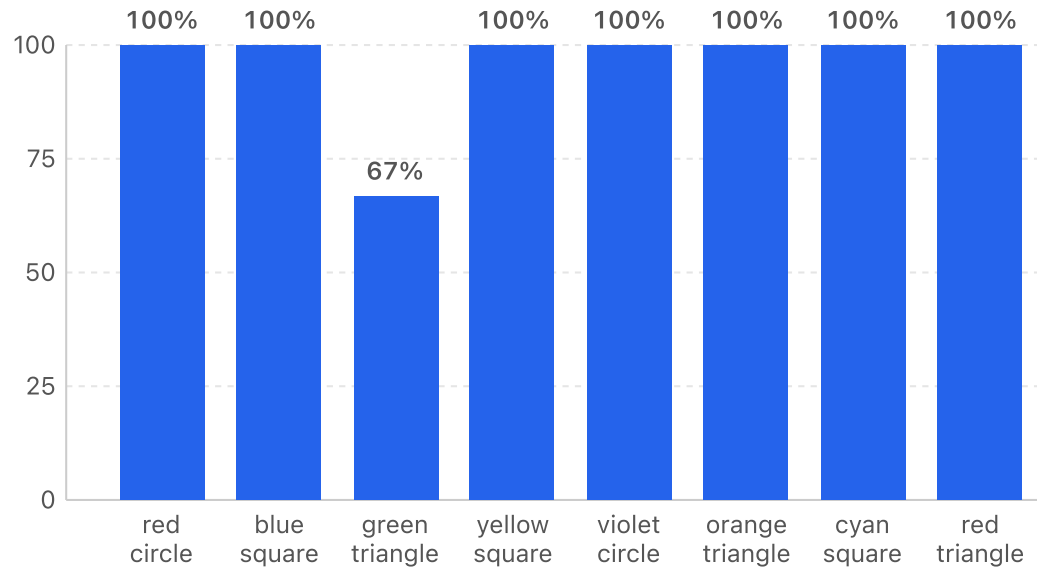


Figure 4. Naming recall for each of eight categories. Three held-out members and three held-out non-members per category, taught from two positives and two counter-examples each. Abstention on non-members was 100% and false namings zero in every category; mean recall was 95.8%.

The rules the substrate induced remain legible at this scale, and remain the conjunctions a person would write down. Table 3 lists them verbatim as recorded by the learning authority.

Category taught	Rule recorded by the substrate	Recall
red circle	<code>circle ∧ vivid_red → red-circle</code>	100%
blue square	more than one hypothesis retained	100%
green triangle	more than one hypothesis retained	67%
yellow square	<code>square ∧ vivid_yellow → yellow-square</code>	100%
violet circle	<code>circle ∧ vivid_violet → violet-circle</code>	100%
orange triangle	<code>triangle ∧ vivid_orange → orange-triangle</code>	100%
cyan square	<code>square ∧ vivid_cyan → cyan-square</code>	100%
red triangle	<code>triangle ∧ vivid_red → red-triangle</code>	100%

Table 3. The eight categories, the rules induced from four labelled images each, and recall on held-out members. Two categories retained more than one consistent hypothesis rather than settling on one; Section 11 reports what naming does in that case.

The cost of the loop is modest and worth stating, because a system that reasons rather than infers is often assumed to be slow. Median times over the run were 28 ms to perceive an image, 1.0 s to admit what was perceived as knowledge, 458 ms to induce a rule from four labelled examples, and 151 ms to name an instance by applying it. The entire study, teaching and testing 149 images across eight categories, completed in 258 s on one machine with no accelerator and no model.

10 What it costs to teach a name

If naming is learned rather than pretrained, the natural question is how much teaching it takes. We measured this directly along the two axes that supervision has: how many labelled examples of the category, and how many counter-examples that discriminate it.

10.1 Labelled examples

Holding the counter-examples fixed, we varied the number of labelled positives from one to four and scored held-out naming each time (Figure 5). With a *single* example the substrate declines to generalise at all, and says so: one example is a case, not a generalisation. This is a deliberate property rather than a limit of the procedure, and it is the same discipline that produces abstention downstream: a system that generalises from one instance has not learned a category, it has memorised an image. With two examples the evidence does not yet pin a single rule down, and naming then proceeds only where the surviving hypotheses agree, which recovers two of the three held-out members (Section 11). From **three** labelled examples the substrate reaches full recall, and a fourth adds nothing. At every setting, abstention on non-members stayed at 100% and false namings at zero: less evidence buys less naming, never more error.

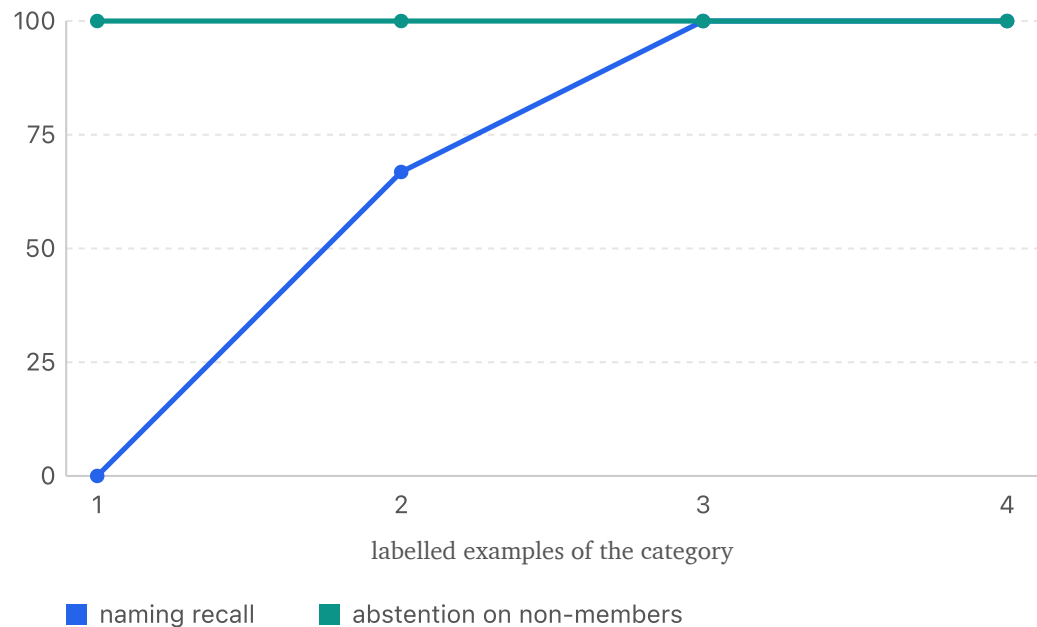


Figure 5. Naming against the number of labelled examples. One example is refused as a case rather than a generalisation; three are enough for full recall. Abstention on held-out non-members is 100% at every setting, and false namings are zero throughout, so the cost of thin evidence is silence rather than error.

10.2 Counter-examples

The second axis is more instructive, because it shows where the honesty in this system actually comes from. Holding the positives fixed at three red circles, all drawn at one size, we varied the number of discriminating counter-examples from none to two and read back both the rule and the behaviour (Figure 6, Table 4).

With no counter-example there is nothing for generalisation to separate, no rule is induced, and the substrate names nothing. With one counter-example (a red square) the rule the substrate records is `circle → category`: the evidence it was given genuinely does not distinguish colour, and the rule states exactly that. It names every held-out member, and it also names a blue circle, because a blue circle is what its evidence licenses. With the second counter-example (a blue circle) the induced rule tightens to `circle ∧ vivid_red → category`, recall stays at 100%, abstention returns to 100%, and false namings return to zero.

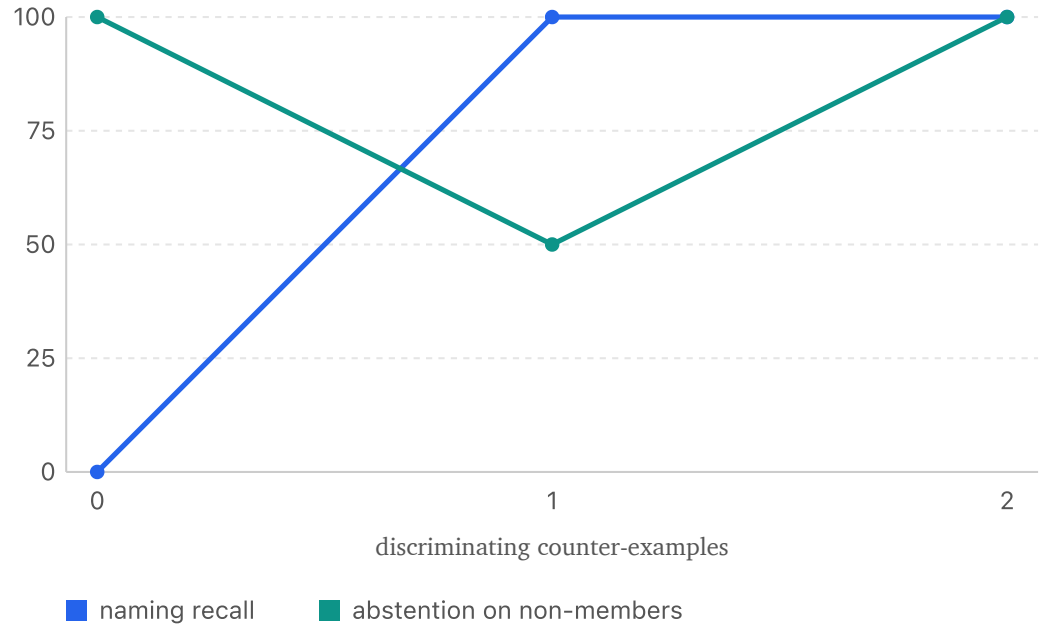


Figure 6. Naming against the number of discriminating counter-examples. Recall rises with the first counter-example; abstention on non-members is restored by the second, which is the one that separates colour from shape. The recorded rule at each setting is exactly as general as the evidence supports.

Counter-examples	Rule recorded	Recall	Abstention	False namings
0	none induced	0%	100%	0
1	circle $\rightarrow$ category	100%	50%	1
2	circle $\wedge$ vivid_red $\rightarrow$ category	100%	100%	0

Table 4. Supervision against behaviour. Three labelled positives throughout, drawn at one size; three held-out members at three other sizes, and two held-out non-members.

The point is not that a single counter-example is insufficient, which is unsurprising. It is that the system’s exposure at that setting is *readable in advance*. The rule is a sentence one can inspect before trusting the system with anything: it says that this category means round, and a person can see immediately that it will therefore accept a blue circle. The corresponding state of a trained network is a set of weights, and the corresponding failure is discovered by encountering it. Here the extent of a generalisation is a property of the record, and teaching one further example both closes the gap and changes the sentence.

11 When the examples do not decide

A learner that keeps a single rule must pick one whenever its examples admit more than one, and a pick made on evidence that does not support it is a guess in a rule’s clothing. This substrate does not pick: induction retains every hypothesis consistent with what it was shown. That raises the question this section answers, which is what *naming* should then do.

We measured it end to end over **416** images on 17 September 2026, through the same reasoning path used everywhere else in this paper. Eight categories were taught twice in each of two arms. In the *clean* arm the two labelled positives differed in size, so the only thing they shared was the category itself. In the *accidental* arm both positives were small, a size the category itself does not require, so they shared a feature by accident, which is what an uncurated teaching set looks like.

The two arms separate completely. All sixteen inductions in the clean arm settled on a single rule. All sixteen in the accidental arm retained exactly two, one built on the shared accident and one on the category’s own features, and the substrate recorded both. For the red circle it held `small → category` and `circle ∧ vivid_red → category` side by side. Both are consistent with every example it was given, and nothing in that evidence prefers either.

11.1 Three policies on identical evidence

Given a set of surviving hypotheses, a system can name in three ways, and we scored all three over the same **288** held-out instances (96 members, 192 non-members) recorded in this study. *Permissive*: name the instance if any surviving hypothesis accepts it. *Strict*: name only in the cases where a single hypothesis survived, and stay silent wherever the evidence was ambiguous. *Unanimity*: name only what every surviving hypothesis accepts. Unanimity is what the substrate does live; the other two are computed from the same recorded hypotheses and reported as counterfactuals on identical data.

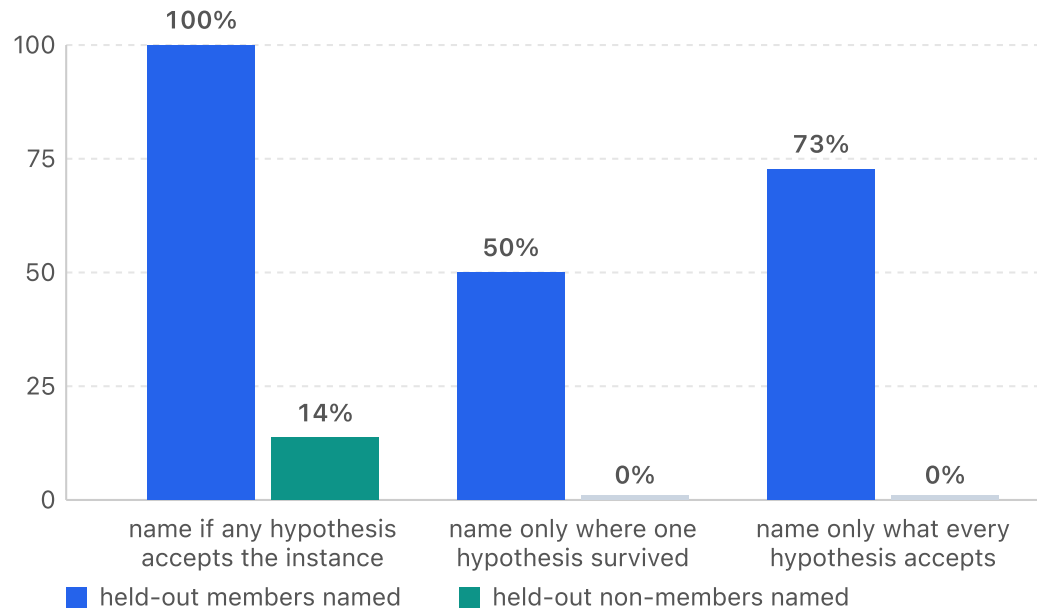


Figure 8. Naming under three policies when the examples do not decide. Identical data and identical recorded hypotheses; 96 held-out members and 192 held-out non-members. The permissive policy names every member and also names 26 non-members. Requiring unanimity names no non-member at all, and still names three quarters of the members, because where the hypotheses agree their disagreement elsewhere does not matter.

Policy	Members named	Non-members named	False namings
Name if any surviving hypothesis accepts	100% (96/96)	13.5% (26/192)	26
Name only where one hypothesis survived	50.0% (48/96)	0% (0/192)	0
Name only what every hypothesis accepts (live)	72.9% (70/96)	0% (0/192)	0

Table 6. Thirty-two inductions, 288 held-out instances. The live policy is the third row; the first two are counterfactual policies scored on the same recorded hypotheses.

Two things follow. The first is that the discipline is doing real work: on evidence a person would consider perfectly ordinary, naming on any consistent hypothesis produces twenty-six confident errors, every one of them a name the system was not entitled to. The second is that abstaining wholesale is not the only alternative. Unanimity recovers nearly twenty-three points of the recall that the strict policy gives up, and it does so without a single false naming, because the instances where the hypotheses disagree are exactly the instances the evidence does not decide. The substrate is silent precisely there, and nowhere else.

11.2 Closing the ambiguity by asking for a case

Silence is the right answer, but it should not be the last word. An undetermined induction also states, in its own terms, the case that would settle it: “an instance that is small but not circle and vivid_red is accepted by one hypothesis and refused by the other”. That is a request for a specific experiment, produced by the learner rather than by a person reading its rules.

One such case eliminates one hypothesis, so the protocol is a loop rather than a single shot: induce, read the request, supply exactly the case it names, induce again. We ran that loop on 17 September 2026 over the sixteen undetermined inductions of the accidental arm, with a cap of four rounds, against a baseline that supplied a randomly drawn stimulus each round instead. Both arms started from the same four labelled images, labelled every supplied example by the same rule, and worked from the same stimulus space of 448 images, each of which was perceived before use and accepted only if perception reported the features the round required. Nothing was assumed from how a stimulus was drawn.

Supplying the requested case closed the ambiguity in **all sixteen**, in a mean of **2.6** rounds: six settled in two rounds and ten in three. Supplying random examples under the same cap closed **one** of the sixteen. The difference is not diligence, it is aim: a random example is usually consistent with every hypothesis standing, and consistency eliminates nothing.

The trace of one category shows what the loop is doing. Taught two small red circles against a medium blue circle and a medium red square, the substrate held two hypotheses and asked for the case that separates them (Table 7).

Round	Case supplied	Hypotheses standing afterwards
0	the four labelled images	$\text{small} \rightarrow \text{cat}$ $\text{circle} \wedge \text{vivid_red} \rightarrow \text{cat}$
1	small blue circle, a non-member asked for: small, but not circle and vivid_red	$\text{circle} \wedge \text{vivid_red} \rightarrow \text{cat}$ $\text{small} \wedge \text{vivid_red} \rightarrow \text{cat}$
2	tiny red circle, a member asked for: circle and vivid_red, but not small	$\text{circle} \wedge \text{vivid_red} \rightarrow \text{cat}$

Table 7. One category’s resolution. Each round eliminates the hypothesis built on an accident of the examples, and the request at each round names the case that does it. The same four starting images, given four random examples instead, still left two hypotheses standing at the cap.

Each round eliminates one hypothesis and the substrate restates what it still needs, which is why the mean is between two and three rounds rather than one. The examples it asks for are the ones a careful teacher would choose and for the same reason: the first separates size from colour and shape, the second separates size from the rest by supplying a member that breaks it.

The request is therefore not a diagnostic message; it is the next example, named in advance. A system that can say which case it needs can be taught deliberately, and a teacher who supplies that case closes the ambiguity in one step rather than by accumulating data and hoping.

12 The operating range of perception

Because the perceptual stage is algorithmic rather than learned, its limits are properties of the measurement and can be mapped exactly rather than estimated. We degraded the stimuli along six axes and recorded, for three shapes at each setting, whether the object was reported at all and whether its shape class and colour family were recovered correctly (Table 5).

Degradation	Holds to	First setting that degrades	What degrades
Additive noise (σ)	$\sigma = 96$	$\sigma = 128$	shape class (1/3); the object is still found and its colour still correct
Gaussian blur (kernel)	$k = 41$	$k = 61$	shape class (2/3); colour unaffected
Rotation	45°	none tested	nothing; shape and colour correct at 10° , 30° and 45°
Desaturation (α)	$\alpha = 0.7$	$\alpha = 0.5$	colour family; the measured dominant colour is no longer the vivid family, which is a true report of the pixels
Occlusion (bar width)	unoccluded	30 px	shape class (1/3); the object is still found and its colour still correct
Apparent size	40 px	20 px	the object falls below the reporting threshold before it is misread

Table 5. Perception under degradation, three stimuli per setting. Colour is the most robust measurement and shape the first to go. These are measurements of the perceptual stage alone.

Two patterns are worth drawing out. Colour survives heavy noise, heavy blur, rotation and partial occlusion, and fails only under desaturation, where the honest answer really is that the pixels are no longer that colour. Shape is the fragile measurement, and it is the one that fails first under every axis that destroys a contour. What this means downstream follows from how naming works rather than from a separate measurement. A learned rule fires on the features perception actually reports, so an object whose shape cannot be resolved supplies no shape to fire on and the substrate declines; an object whose shape is misreported supplies the wrong one, and the rule fires or fails according to that feature rather than to the system’s confidence. The

boundary above is therefore a boundary on what can be named at all, and it is stated in the same measured terms as everything else in this paper.

The apparent-size row is a reporting threshold rather than a resolution limit, and it is worth separating the two. The faculty ignores blobs below a minimum share of the frame, by default 1%, so that texture and compression artefacts are not admitted as objects. Below that share an object is not misread, it is not reported. Lowering the threshold to 0.1% recovers objects down to 0.13% of the frame (Figure 7). Colour is reported correctly at every size at which the object is reported at all; the shape class is correct down to 0.89% of the frame and is not dependable below it, which is the same ordering seen under every other degradation. The limit is therefore a setting with a stated trade-off, not a boundary of the approach.

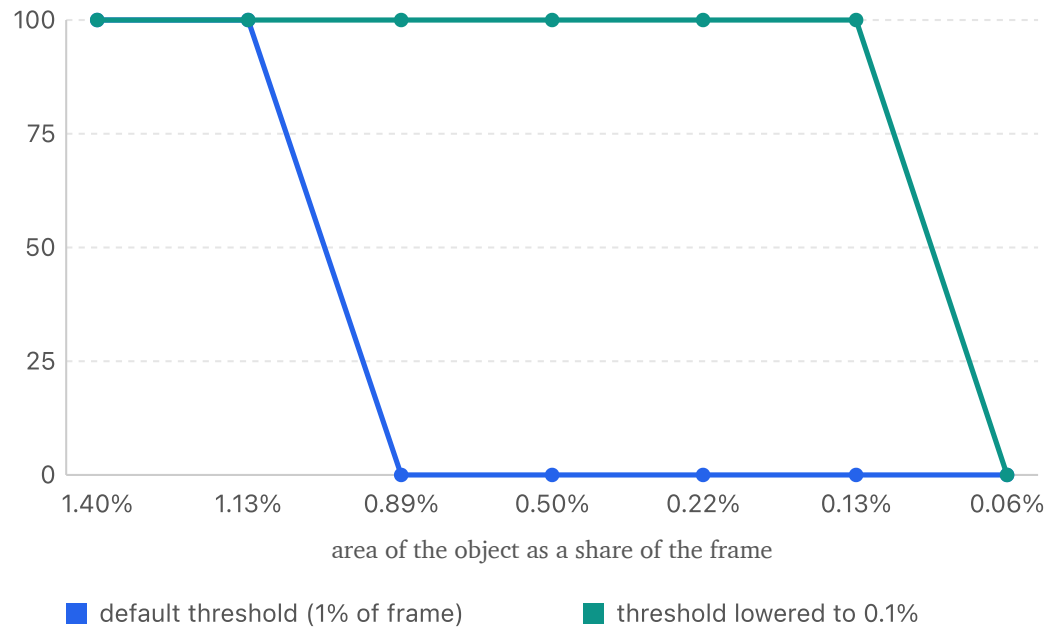


Figure 7. Whether an object is reported, against its share of the frame. At the default minimum area of 1% of the frame, objects below that share are not admitted as objects. Lowering the threshold to 0.1% admits them down to 0.13% of the frame. Colour is correct wherever the object is reported; the shape class is correct down to 0.89% of the frame.

13 Discussion and limitations

This is a proof of concept, and its scope should be read plainly. The categories are visually simple and the stimuli controlled; the point of that control is to make perception, learning and abstention separately measurable and the ablation clean, not to claim performance on natural imagery. The naming demonstrated here is learned from a few labelled examples over features the perceptual stage recovers; it is not open-vocabulary recognition of arbitrary categories, and it inherits the reach of classical perception, which recovers

structure and specific known instances but does not, by itself, put a general name to a novel natural object. That last step is precisely the one the substrate is meant to learn, and extending the feature vocabulary and the range of learnable categories is the natural next stage of the work.

The larger study sharpens the scope rather than widening it. Eight categories over 149 images is still a controlled setting: the stimuli are drawn, the feature vocabulary is the one the perceptual stage recovers, and the categories are conjunctions within it. What the scale buys is confidence that the behaviour reported in Section 5 is a property of the loop and not of three convenient categories, and that the honesty holds where the accuracy does not: across every study in this paper, including the deliberately under-supervised and deliberately ambiguous conditions, the substrate produced no name it could not ground in a rule it had learned, and one false naming in total, in the single condition whose evidence licensed it and whose recorded rule said so in advance.

The supervision study also gives a concrete figure for what this architecture costs to teach, which the model-based route pays once and globally and this route pays per category: three labelled examples and two discriminating counter-examples were enough to determine a rule, and four labelled images determined one in six of the eight categories taught at scale. That is cheap in absolute terms and expensive relative to a pre-trained vocabulary of a thousand classes, and it is the honest trade. What is bought in return is a naming ability that is attributable to specific evidence, revisable by supplying more of it, legible before it is trusted, and silent where it has none.

What the proof of concept establishes is the architecture: that a substrate with no perceptual model can nonetheless see, by pairing algorithmic perception with its own inductive learning, and that doing so buys a property the model-based route does not: the system knows when it does not know. The naming ability is learned, attributable, revisable and legible; its failures are abstentions rather than confident errors; and none of it depends on a pretrained backbone. Where a conventional pipeline would place a frozen network to turn structure into names, this substrate places a learner it already had.

14 Conclusion

We set out to give sight to a system that carries no neural network, by taking seriously the idea that vision is structure perceived and meaning learned. The demonstration is complete within its scope: real images are perceived into structure with deterministic algorithms; a few labelled examples teach the substrate a rule that names a category; new instances are named by applying that rule, and unlearned ones are declined. On a controlled evaluation the substrate named held-out instances at full recall with no false namings and no model calls, and an ablation showed the ability to be carried entirely by what it learned, its absence surfacing as honest abstention. Repeating the study across eight categories and 149 images left that picture unchanged; measuring the supervision a name requires put its price at a handful of labelled images; and where those images left more than one reading open, the substrate named only what all of them agreed on and

said which case would settle the rest. A cognitive substrate can therefore supply the semantic half of vision itself, in a form that can be questioned, corrected and audited.

References

- [1] Dominion Labs. Structure Before Meaning: giving a cognitive substrate sight. Dominion Labs Research, 2025.
- [2] A. Krizhevsky, I. Sutskever, G. E. Hinton. ImageNet classification with deep convolutional neural networks. In *NeurIPS*, 2012.
- [3] A. Radford et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.
- [4] P. F. Felzenszwalb, D. P. Huttenlocher. Efficient graph-based image segmentation. *International Journal of Computer Vision*, 59(2):167–181, 2004.
- [5] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, A. W. M. Smeulders. Selective search for object recognition. *International Journal of Computer Vision*, 104(2):154–171, 2013.
- [6] J. Matas, O. Chum, M. Urban, T. Pajdla. Robust wide-baseline stereo from maximally stable extremal regions. *Image and Vision Computing*, 22(10):761–767, 2004.
- [7] A. Kirillov et al. Segment Anything. In *ICCV*, 2023.
- [8] Harnessing vision foundation models for high-performance, training-free open-vocabulary segmentation. *arXiv:2411.09219*, 2024.
- [9] R. Szeliski. *Computer Vision: Algorithms and Applications*. Springer, 2010.
- [10] D. G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2):91–110, 2004.
- [11] G. D. Plotkin. A note on inductive generalization. *Machine Intelligence*, 5:153–163, 1970.
- [12] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.
- [13] R. Reiter. On closed world data bases. In H. Gallaire, J. Minker (eds.), *Logic and Data Bases*, Plenum Press, 1978.

Dominion Labs Research & Development · September 8, 2026. A proof of concept. All figures are from live, model-free runs of the substrate over real images; the evaluation induces its rules from labelled examples and is scored only afterward, never with answers supplied to the reasoner.